Muskan Dhamuria
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EDUCATION

Singapore Management University **Aug 2024 – May 2028**
Bachelor of Science (Information Systems) – Double Major in Product Development and Cybersecurity

- Expected Cum Laude (Distinction)
- Activities: SMU Women in Tech, SMU .Hack, UWS STEMentorship, SMU AI Club
- Teaching Assistant: IS112 Data Management, CS440 Cybersecurity Foundations
- Leadership: Events Deputy of SMU Women in Tech

WORK EXPERIENCE

CPF Board **Jan 2027 – Jun 2027**
Cloud Security and TechOps Intern

- Support mission-critical cloud infrastructure and security platforms, including troubleshooting, performance optimisation, and platform maintenance.
- Collaborate with technical and business stakeholders to identify use cases for IT operations optimisation and automation.
- Develop monitoring solutions and security controls for cloud environments to support infrastructure security and resilience.
- Develop technical documentation, standard operating procedures, and knowledge resources to improve operational clarity and consistency.
- Contribute to cloud operations and automation initiatives using scripting, infrastructure-as-code, and query-based analysis.

SP Group **Jul 2026 – Dec 2026**
AI Automation Engineer

- Migrated legacy IQ Bot document-processing workflows to modern AI-powered Document Automation solutions, improving maintainability and scalability.
- Designed and optimized intelligent document-processing pipelines to enhance data extraction accuracy and reduce manual processing effort.
- Developed Agentic AI workflows for intelligent decision-making, task orchestration, and end-to-end business process automation.
- Collaborated with business stakeholders and cross-functional teams to gather requirements, design AI-driven automation solutions, and support deployment.
- Leveraged technologies including Automation Anywhere, Document Automation, OCR, Generative AI, and Agentic AI to streamline enterprise workflows.

Science Centre Singapore **Jan 2022 – Feb 2022**
Tech Intern

- Developed an interactive gamified 'Young Engineer Trail' and online modules for the 'Young Digital Fabricator' badge program.
- Took full ownership of project delivery following teammate withdrawal, ensuring 100% on-time completion of deliverables.

AWARDS & COMPETITIONS

- **Finalist, Morgan Stanley Code to Give Hackathon 2026** - Selected as one of 50 students across Singapore to develop Passion Hub, a full-stack platform that streamlines event planning, volunteer coordination, participant registration, QR attendance, and post-event administration for a local non-profit. Built an AI Event Copilot that generates workflows, recommends actions, and automates reports, certificates, and social media content. Integrated WhatsApp for automated communications.
- **Finalist, JP Morgan Chase Code for Good Hackathon 2026 (Incoming)**
- **Featured Project, Garena AI Build Challenge 2026 -** Selected as one of 39 featured projects for Emergent, a four-player pixel adventure powered by an AI Game Master that analyses player behaviour in real time, assigns personalised roles, dynamically evolves the game world, and generates adaptive cooperative finales.
- **Most Feasible Solution, Hack For Cities Hackathon 2026** - Developed Urbanix, an AI-powered urban policy optimisation platform using multi-objective optimisation and Pareto frontier analysis.

- **Honourable Mention, FinTech Innovators' Hackathon 2026** - Developed FinTellect, an AI-powered financial health platform that uses a Digital Financial Twin, predictive simulations, behavioural analysis, and an AI Wealth Copilot to assess and improve users' financial resilience.

PROJECTS

ESMOS Cloud-Native Healthcare Platform

- Designed and deployed cloud-native enterprise applications on Azure Kubernetes Service (AKS) using Docker, ArgoCD GitOps, and CI/CD pipelines, achieving 99.9% uptime. Implemented TLS-secured ingress routing with NGINX Ingress and cert-manager, container orchestration, persistent storage provisioning, monitoring/log analytics, and automated deployment workflows for scalable microservice-based healthcare systems.
- Technologies: Kubernetes, Docker, Azure AKS, Azure Container Registry, ArgoCD, GitHub Actions, NGINX Ingress, cert-manager, PostgreSQL, Azure Files, Azure Managed Disks

Web Vulnerability Scanner

- Built an automated web vulnerability scanner capable of detecting SQL Injection, reflected XSS, and missing security headers.
- Implemented a BFS-based crawler, risk-based vulnerability scoring, and multi-format reporting (HTML, JSON, CSV) to support security assessments.
- Technologies: Python, Requests, BeautifulSoup, Web Crawling (BFS), Application Security Testing (SQLi, XSS), Security Headers, Data Reporting (JSON, HTML, CSV)

Koryori Hayashi - Smart Restaurant Intelligence System

- Designed and developed a full-stack digital transformation ecosystem integrating online ordering, loyalty management, operational analytics, and queue management systems.
- Implemented a machine learning-powered digital twin to simulate customer demand, inventory usage, and business performance under different operational scenarios, supporting data-driven decision-making.
- Achieved Higher Distinction for the project during coursework.
- Technologies: React.js, TypeScript, Tailwind CSS, Node.js, Supabase, Recharts, Gemini API, Machine Learning, Thompson Sampling, Digital Twin Simulation

Wingbound - Minecraft Mod

- Developing a custom Minecraft mod featuring interactive dragon companions with feeding, player-following AI, bonding, growth progression, and rideable entities.
- Technologies: Java, Fabric API, Minecraft Modding, Gradle, Object-Oriented Programming, Entity AI, Pathfinding, Custom Animation, JSON, Blockbench, GeckoLib.

CERTIFICATIONS

- Google Cybersecurity Professional Certificate
- Google Data Analytics Professional Certificate
- Google AI Professional Certificate
- Heicoders Academy AI100: Python Programming and Data Visualisation

SKILLS

- **Programming Languages:** Python, Java, JavaScript, TypeScript, PHP, Bash, SQL
- **Web & Software Development:** React.js, Vue.js, Flask, Node.js, Firebase, HTML, CSS, Bootstrap, Tailwind CSS, Automation Anywhere
- **Cloud & DevOps:** Azure, Docker, Kubernetes, ArgoCD, GitHub Actions
- **Cybersecurity:** Kali Linux, Metasploitable, Web Application Security Testing, Vulnerability Assessment
- **Databases & Analytics:** MySQL, Pandas, Power BI, Tableau, Power Query, Matplotlib, Folium, Microsoft Excel
- **Tools & Design:** Figma, Signavio